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Color congruence on Instagram: the role of flow and aesthetic experience

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Abstract

Purpose – Social media has emerged as a prominent platform for marketers and brands to disseminate brand-related information. This study aims to investigate the impact of color congruence between themes and background on marketing effectiveness, focusing specifically on Instagram.

Design/methodology/approach – A laboratory experiment was conducted to investigate how color congruence between themes and background in brand posts influences flow and aesthetic experience, subsequently affecting marketing communication parameters such as brand attitude, visit intention, and eWOM on Instagram. Moreover, Adidas Originals was selected as the focal brand, and blue and white color was chosen as the primary color palette for the experimental material.

Findings – This study demonstrates that color congruence, regardless of brand layout or post, significantly influences flow and aesthetic experience, subsequently affecting marketing effectiveness.

Originality/value – This study contributes to the theoretical understanding of congruence theory and social media marketing, providing valuable insights for brands to enhance their communication through photographs and effectively manage their official Instagram accounts.

Keywords Instagram, Color congruence, Flow experience, Aesthetic experience, Marketing effectiveness

Paper type Research paper

1. Introduction

Owing to its visual-centric nature, Instagram has emerged as the predominant social media platform among the young (Jiang *et al.*, 2019). Instagram consistently refines its visual elements, infusing them with nuanced connotations and updating its system functions, making it highly adept at capturing consumers' attention (Virtanen *et al.*, 2017; Wang and Weng, 2023). To effectively engage consumers, marketers must craft visual images capable of capturing attention, facilitating users to scroll through content for cognitive processing (Cheung *et al.*, 2020; Chiu, 2022). Consequently, researchers (Agnihotri *et al.*, 2023; Zappavigna, 2016) have increasingly delved into understanding how heuristic cues influence users' attitudes and beliefs concerning brand posts in online environments, forming the basis for this study.

In the context of information overload, users often process online information relying on visual cues that convey a brand's values, identity, and personality (Chiu *et al.*, 2017). When consumers interact with the visual elements in product advertising, they identify these components, constructing an impression of both the brand and the product (Adaval *et al.*, 2019). Notably, consumers' rapid perceptions of a brand's image influence their attitudes toward advertising and shape their subsequent views of the brand (Parris and Guzmán, 2023; Zappavigna, 2016). Against this backdrop, the current study investigates whether visual cues within brand-related posts on Instagram influence consumers' attitudes and behaviors, specifically addressing the question: *Does the color composition in brand posts affect marketing effectiveness?*

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This study delves into the impact of color cues among visual elements on shaping consumers' experiences and perceptions of brand posts. Existing research indicated that consumers typically prioritize color cues when browsing brand posts (Sundar and Kellaris, 2017). These cues serve as informative signals, offering insights into unfamiliar brands, conveying characteristics and images, and strategically triggering heuristics in digital media (Chiu *et al.*, 2017; Zailskaitė-Jakštė *et al.*, 2017). Notably, colors play a crucial role in influencing flow experience and perceptions evolving based on the context in which colors are experienced. Farace *et al.* (2020) highlight that a congruent color scheme diminishes the perceived distance between the brand and the consumer, consequently enhancing the brand's perception. Congruent visual patterns offer a systematically complete image that is more accessible for viewers to recognize and comprehend, effectively engaging the mind compared to incongruent patterns (Kim and Jang, 2018; Wong *et al.*, 2019). As viewers seamlessly associate the brand and its products, this heightened color congruence contributes to a positive flow experience, influencing consumers' evaluations positively (Jiang *et al.*, 2019; Schwarz *et al.*, 2021).

Moreover, the emphasis on visual aesthetics in previous studies has been overshadowed by a predominant focus on the usability features of websites. Yet, it is crucial to acknowledge that aesthetics play a pivotal role in the success of a website, influencing its capacity to capture the attention of potential users and deliver an engaging interactive experience (Agnihotri *et al.*, 2023). Scholars increasingly recognize that design elements, including color, graphics, and layout, substantially influence not only the usability and enjoyment of information systems but also overall work quality. Therefore, the primary motivation behind this study is to unravel how color congruence enhances the persuasiveness of brand posts, addressing a critical aspect often overlooked in existing literature.

The primary aim of this study was to investigate whether color congruence fosters flow and aesthetic experiences, thereby extending the communication of brand-related information from textual to visual dimensions. This study posited that color congruence between background and themes plays a pivotal role in influencing brand layouts/posts on Instagram. This study employed an experimental approach to replicate the Instagram interface and content of official brand accounts, incorporating four conditions of color (congruence vs incongruence) and type (brand posts vs brand layout). To assess the model and test the hypotheses, this study simulated the scenario of an official Instagram account, utilizing "Adidas Originals" as an illustrative example to determine the impact of color congruence on consumers' flow and aesthetic experiences, subsequently enhancing marketing effectiveness (brand attitude, visit intention, and eWOM). The findings contributed to a more profound understanding of the influence of color cues in brand posts on Instagram. As a result of this understanding, marketers can craft marketing content that not only engages their audience but also persuades them effectively.

2. Literature review

2.1 Instagram and brand posts

As social media platforms have evolved into visually-driven spaces, allowing users and brands to share images, photos, and videos along with accompanying information (Lee *et al.*, 2015), the motivations driving Instagram use are identified as sharing life experiences and fostering social interactions (Agnihotri *et al.*, 2023). Instagram significantly influences consumers' decision-making processes, with hashtag searches facilitating the discovery of desired brands and products (Zappavigna, 2016). Recognizing its potential, brands have established official Instagram accounts to engage potential consumers through creative and visually compelling content (Cheung *et al.*, 2020).

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Despite the acknowledged importance of visual elements in shaping brand perceptions (Zailskaitė-Jakštė *et al.*, 2017), research on visual cues, particularly the impact of color, remains limited. The color effect, notably, has received minimal attention, despite the fact that consumer perceptions of brand content are influenced by colors; for instance, a website's background color can shape Taiwanese consumers' price perceptions (Hsieh *et al.*, 2018). While previous studies have recognized the significance of Instagram as a platform for shaping consumers' moods, impressions, and brand images (Parris and Guzmán, 2023; Virtanen *et al.*, 2017), a gap persists in understanding the role of color composition in influencing how users evaluate brand messages on Instagram. This study aims to address this gap by focusing on a specific color cue on the Instagram platform: congruence.

2.2 Color congruence, flow experience and aesthetic experience

Congruence is a state that denotes the relatedness, identity, and suitability of two or more objects placed together (Belanche *et al.*, 2017). According to Lee and Schumann (2004), in a state of congruence, users can integrate existing knowledge structures and frameworks when evaluating information. This integration facilitates a smoother cognitive processing, allowing users to process information more efficiently and with reduced cognitive effort. Additionally, congruent information is more likely to be recognized, recalled more promptly and frequently, retained for longer periods, and elicit a more profound aesthetic experience and positive attitude compared to incongruent information (Lee and Kim, 2022; Wong *et al.*, 2019).

Information congruence with users' current knowledge structures is more likely to be accepted, while information incongruent with their existing knowledge schema is likely be filtered out, leading to the congruence effect (Crutzen *et al.*, 2012). Moreover, images with congruent colors facilitate smooth cognitive processing, as they are easily recalled in users' minds (Ettis, 2017; Kim and Jang, 2018), resulting in a flow and aesthetic experience. In contrast, the introduction of information incongruence into the brain of the viewer, due to its mismatch with the existing schema, hampers cognitive processing and induces discomfort (Lee and Kim, 2022). During this period, users focus on cognitive processing as a means to enhance attention to the inconsistent information. If the inconsistent information is successfully resolved, it leads to the viewer's psychological satisfaction, thereby increasing subsequent goodwill (Mazodier and Quester, 2014).

In contrast to previous studies, this study specifically examines the impact of color congruence and incongruence on images and photos shared on Instagram. Color congruence, in this context, refers to the harmonious alignment between the colors of themes and their backgrounds. To illustrate, consider a scenario where the background color seamlessly matches the theme color in brand posts. This congruence facilitates quicker processing of visual information by consumers, enhances their understanding, and triggers a positive flow and aesthetic experience. In stark contrast, incongruent colors between the background and themes in brand posts disrupt users' expectations, leading them to invest more time in comprehension, thereby increasing overall attention (Ettis, 2017; Wong *et al.*, 2019). That is, a brand's Instagram post with a well-executed color congruence may reduce the cognitive effort required from viewers, leading to a more enjoyable and seamless browsing experience. On the contrary, posts with incongruent colors might prompt viewers to pause and analyze the content more intensely. Thus, this study asserted that the application of color congruence in brand layouts/posts significantly enriches the brand ambiance, fostering a positive association with the flow and aesthetic experience among viewers. The following hypotheses are posited based on this premise:

H1a. Higher color congruence on brand post/layout is positively associated with flow experience on Instagram.

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H1b. Higher color congruence on brand post/layout is positively associated with aesthetic experience on Instagram

2.3 Flow experience, aesthetic experience and marketing effectiveness

Stiff and Mongeau (2016) stated that marketing effectiveness as the measure of whether an advertising persuades consumers after its display. According to previous study, the flow experience has been identified as a factor that improves message evaluations and fosters a preference for messages processed at faster speeds (Hyun *et al.*, 2022; Kim and Jang, 2018). When advertising triggers a flow experience in consumers, it tends to enhance their acceptance of the advertising, capture more attention, and ultimately elevate the effectiveness of marketing communication (Cuevas *et al.*, 2021). Numerous studies have also suggested that the flow experience, occurring post advertising viewing, significantly influences consumers' subsequent behaviors (Cuevas *et al.*, 2021).

The aesthetic experience is a holistic perception that encompasses both design principles and individual elements (Kim *et al.*, 2020). The creation and perception of any object involve specific design elements such as shape, color, and shade, as well as principles like unity, contrast, balance, and proportion. An aesthetic experience captures users' attention, fostering engagement and immersion in an activity (Seifert and Chattaraman, 2020). Furthermore, it involves a cognitive process where a viewer elucidates, intensifies, and interprets events in a visual environment. In this context, color emerges as one of the most impactful factors influencing the aesthetic experience. Alterations in colors on a webpage, such as background color, are straightforward and adaptable, yet they wield a substantial visual influence on consumers, given their capacity to fill the entire computer screen.

The subsequent behaviors discussed in this study can be broadly categorized into three major aspects: brand attitude, visit intention, and eWOM. Brand attitude is defined as consumers' general opinions about a brand, shaped by their beliefs and emotions (Chiu, 2019). Specifically, these opinions are formed through consumers' evaluation of various aspects of the brand. For example, consumers are more likely to make a purchase when they experience positive emotions and flow in relation to a brand, such as feelings of pleasure, enjoyment, and satisfaction (Chiu *et al.*, 2017). That is, conceptual fluency refers to the ease in processing the meaning of the object through the availability of appropriate mental concepts. The positive affect reactions stemming from flow experience may serve as a basis for judgment and contribute to the stimulation of a positive attitude (Hyun *et al.*, 2022; Sohn, 2017).

Additionally, aesthetic experiences can influence online consumers' perceptions by eliciting affective responses and may impact the value of the shopping process by enhancing consumer shopping performance (Kim *et al.*, 2020). Indeed, product image triggers associations in consumers' minds related to utilitarian goals, serving as a cognitive anchor that enhances brand attitude. Given these considerations, this study suggested that higher color congruence on brand post/layout is positively associated with flow and aesthetic experience, which may further impact brand attitude on Instagram, leading to the formulation of the following hypotheses:

H2a. Higher flow experience is positively associated with brand attitude on Instagram.

H2b. Higher aesthetic experience is positively associated with brand attitude on Instagram.

One crucial metric employed to assess webpage performance is visit intention, providing insights into consumers' likelihood to explore a webpage and potentially influencing their subsequent behavior based on the displayed information (Chiu *et al.*, 2017). For example, when consumers encounter visually appealing website layouts and rich images, it can evoke

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a flow experience, fostering an immersive and engaging interaction with the content (Cuevas *et al.*, 2021; Yang *et al.*, 2020). Consider the scenario where a fashion brand employs vibrant and cohesive color schemes, creating an aesthetically pleasing and visually engaging Instagram feed. As users scroll through the brand's posts, experiencing a seamless and visually pleasing flow, they are more likely to be motivated to actively search for the brand. This positive flow experience, triggered by the layout design, can effectively prompt clicks on the brand page, leading to increased visit intention. Therefore, this study suggests that flow experiences not only enhance users' interest in seeking brand-related content but also elevate their intention to visit brand accounts on Instagram.

Furthermore, aesthetic experience is defined as a subjective encounter with visual perception, encompassing the interpretation of pleasantness and involves the decoding of design along with its symbolic meaning (Seifert and Chattaraman, 2020). Previous research has indicated that aesthetic experience contributes to conveying a distinct and clear image, assisting users in accomplishing their goals during website visits (Kim *et al.*, 2020). In the context of Instagram, when consumers' flow and aesthetic state are stimulated by brand content, such as higher color congruence on brand post/layout, it is likely to elevate their visit intention. Building on these observations, the following hypotheses are proposed:

H3a. Higher flow experience is positively associated with visit intention on Instagram.

H3b. Higher aesthetic experience is positively associated with visit intention on Instagram.

eWOM pertains to communication between communicators and recipients who perceive messages as non-commercial, even when discussing topics related to a brand, product, or service (Luarn *et al.*, 2016; Wang and Weng, 2023). Extensive research by Williams *et al.* (2011) and Yu *et al.* (2018) indicated the impact of images in representing consumer interests, highlighting the role of visual congruence, particularly in visual themes, in influencing purchase behavior and eWOM. Visual congruence, marked by consistent color schemes and themes, enhances accessibility and comprehension, thereby stimulating cognitive processes effectively (Wong *et al.*, 2019). Moreover, meaningful visual design incorporating product associations can elevate aesthetic judgments by effectively conveying essential conceptual information that primes the product (Seifert and Chattaraman, 2020). In the context of our study, we posited a positive association between higher color congruence on brand posts/layouts and users' flow and aesthetic experience on Instagram. For instance, consider a fashion brand consistently employing visually congruent color schemes and themes in its Instagram posts, creating a cohesive and visually appealing feed. As users engage with this content, experiencing a seamless and aesthetically pleasing flow, they are more likely to find the brand visually engaging and stimulating. Therefore, this study indicated that:

H4a. Higher flow experience is positively associated with eWOM on Instagram.

H4b. Higher aesthetic experience is positively associated with eWOM on Instagram.

3. Method

3.1 Research model

The purpose of this study was to examine the impact of color congruence in brand posts on flow, aesthetic experience, and subsequent marketing communication outcomes. To accomplish this goal, an experimental design was implemented to establish a controlled environment, mitigating the influence of extraneous factors on marketing effectiveness. A laboratory experiment was conducted to investigate how color congruence between themes and background in brand posts influences flow and aesthetic experience,

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subsequently affecting marketing communication parameters such as brand attitude, visit intention, and eWOM on Instagram. Thus, this study adopted a 2 (color: congruence vs incongruence) x 2 (type: brand post vs brand layout) between-subjects experimental design. The research model was illustrated in Figure 1.

3.2 Research procedures

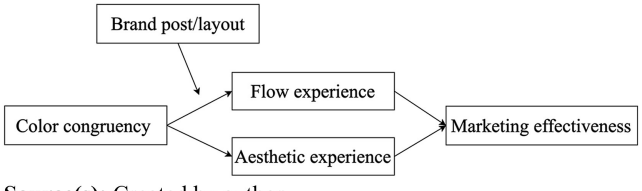
To emulate the genuine experience of interacting with brand posts on Instagram, Adidas Originals was selected as the focal brand for this study. Participants were kept uninformed of the study's objectives until they had completed all phases. Prior to entering the experimental conditions, participants were assigned a task and informed that they needed to explore either an Adidas Originals brand layout or a brand post on Instagram, which constituted the primary content of the experiment.

Participants were randomly assigned to one of four experimental conditions. Subsequent to exposure to the experimental content, they were instructed to complete a questionnaire covering various dimensions such as flow experience, aesthetic experience, brand attitude, visit intention, eWOM, and demographic information. Upon finishing the experiment, participants were briefed on the research's purpose. On average, participants spent approximately five minutes on the experiment.

3.3 Experimental material

3.3.1 Choosing the specific brand official account. To establish an experimental environment that closely mirrors consumers' typical browsing experiences on Instagram, the official account of "Adidas Originals" was chosen as the test brand and content source. "Adidas Originals" stands out as one of the most popular sports brands in Taiwan, known for its emphasis on creativity and popularity as design indicators. In addition, it is plausible that Adidas Originals employs a series of coordinated color schemes in its marketing activities, aligning well with the thematic focus of this study. Consequently, Adidas Originals was identified as the experimental brand, and the experiment was designed around consumers browsing Instagram brand accounts on smartphones.

3.3.2 Manipulating the brand posts. For the experiment, four versions of brand posts were generated as experimental stimulus to closely replicate authentic content on Adidas Originals' official account. Each post in this study comprised specific sections: the brand logo, the themes (featuring models and products) with specific color, and the background color. The manipulation of stimulus content involved utilizing actual posts from the official account, with an emphasis on maintaining the original format of a brand layout or brand post to prevent confusing or misdirection among participants. To ensure the participants' focus was not disrupted, the brand post remained static. Therefore, apart from changes in the background color and themes color, all other elements of the stimulus posts were held constant. This included the format, posting time, number of likes, as well as font and size.



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Figure 1.
Research model

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3.3.3 Manipulating the colors on brand layout/post. Since colors carry product-related meanings and associations, maintaining consistent denotation levels can draw in consumers (Aslam, 2006). Adidas Originals' classic color scheme, predominantly blue and white, was chosen as the primary color palette for the experimental material. Moreover, blue holds universal meanings (Amsteus *et al.*, 2015) and is a color rarely associated with negative user reactions (Singh, 2006). Consequently, these two foundational colors were deemed suitable for investigating the color effect in this study.

Specifically, the blue and white backgrounds, along with the featured products in Instagram posts, were manipulated based on content from official Adidas Originals accounts. Four conditions were tested in this study: (a) In the brand post with color congruence condition, we modified the photo to include a model wearing the products in blue color, and the background color was also largely blue with a little white color. (b) In the brand post with color incongruence condition, we used the same photo but modified interspersed various colors (such as yellow, red, purple, etc.) on themes and background in order to strengthen the feeling of incongruence. Therefore, the main differences between condition (a) and (b) were the color of themes and background.

To further clarified whether the effect of color congruence on brand layout, this study simulated the realist Adidas Originals brand layout on Instagram. (c) In the brand layout with color congruence condition, this study used nine brand-related photos with similar blue/white color. (d) In the brand layout with color congruence condition, we used same photo as condition (c) and modified the color in each photo which makes photo become the colors diversity on themes and background. Table 1 shown that the types and elements of the photos are largely the same, to minimize interference between the experiment and other factors. The colors were also varied to strengthen the sense of incongruence.

Before the formal experiment, 30 participants were recruited to confirm that the states are successfully manipulated in this study. Each item was composed of a 7-point Likert scale anchored at each end by a pair of phrases that expressed negative and positive feelings. The congruence scale was adapted from Sujan and Bettman (1989) and ranged from 1 (strongly disagree) to 7 (strongly agree) (Cronbach's $\alpha = 0.82$ in this study).

3.4 Measurements

The questionnaire used in this study was revised based on scale items from previous studies. The participants responded to the statement 'I feel (this post) is . . . ' by using 7-point Likert scale ranging from strongly agree to strongly disagree. The measurement items for flow experience, with four reflective dimensions (time distortion, immersion, control, and enjoyment), were adapted from Agarwal and Karahanna (2000) (*Cronbach's* $\alpha = 0.83$). Aesthetic experience were adapted from Cai and Xu (2011) (*Cronbach's* $\alpha = 0.81$). Brand

Variable	Type	color	n	M	SD	F	*p*
Flow experience	layout	congruency	68	5.19	1.11	10.69	<0.001*
		incongruency	63	4.74	1.44		
	post	congruency	79	4.89	1.13		
		incongruency	70	4.02	1.38		
Aesthetic experience	layout	congruency	68	4.88	0.99	17.28	<0.001*
		incongruency	63	3.97	1.14		
	post	congruency	79	4.34	1.28		
		incongruency	70	3.46	1.31		

Note(s): N = number of participants; M = mean; SD = standard deviation; *t* = *t*-value; *p* = *p*-value
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Table 1.
The impact of color congruence on processing fluency

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attitude with five items (*Cronbach's* $\alpha = 0.91$) and visit intention with four items (*Cronbach's* $\alpha = 0.87$) were adopted from the scale developed by Heo *et al.* (2023) and Zarantonello and Schmitt (2013). In addition, eWOM was determined by four items from the scale developed by Yu and Ko (2021) (*Cronbach's* $\alpha = 0.85$ in this study). (The specific items in each scale are listed in the Appendix).

3.5 Participants

The purpose of this study was to verify the impact of color congruence on Instagram, so we recruited volunteers who had experience using Instagram to participate in the experiment. Participants were randomly assigned to one of the four experimental groups. After excluding invalid subjects, a total of 280 participants took part in the experiment, comprising 101 males and 179 females. The average age was 20.44 years (SD = 2.41). The majority of participants fell within the 18–22 years old and held a university degree. Participants had an average Instagram usage experience of 7.11 years (SD = 2.23), with an average daily Instagram usage time of 1.65 h per day (SD = 1.94).

4. Results

Exploratory factor analysis and reliability analysis were conducted using the SPSS 25.0 program. Principal component factor analysis and varimax rotation were employed as methods of factor extraction, selecting factors with an eigenvalue of 1.0 or above and a factorial load of 0.5 or above. Internal consistency was assessed through the Cronbach's alpha value in the reliability analysis.

In this study, we hypothesized that brand posts/layouts with color congruency between themes and background would positively associate with flow experience on Instagram. According to the ANOVA results, color congruency on brand layouts ($M = 5.19$, $SD = 1.11$) and brand posts ($M = 4.89$, $SD = 1.13$) significantly affects flow experience than color incongruency situations ($F = 10.69$, $p < 0.001$), supporting H1a. Moreover, color congruency on brand layouts ($M = 4.88$, $SD = 0.99$) and brand posts ($M = 4.34$, $SD = 1.28$) significantly affects aesthetic experience than color incongruency situations ($F = 17.28$, $p < 0.001$), supporting H1b (see Table 1 for details).

H2–H4 suggested that higher flow experience is positively associated with marketing effectiveness on Instagram. The results of regression analysis indicated that higher level of flow ($\beta = 0.60$, $p < 0.001$) and aesthetic ($\beta = 0.32$, $p < 0.001$) experience were positive related with brand attitude ($\Delta R^2 = 0.43$), supporting H2a and H2b. Moreover, the results also indicated that higher level of flow ($\beta = 0.43$, $p < 0.001$) and aesthetic ($\beta = 0.32$, $p < 0.001$) experience were positive related with visit intention ($\Delta R^2 = 0.49$), supporting H3a and H3b. The higher level of flow ($\beta = 0.35$, $p < 0.001$) and aesthetic ($\beta = 0.35$, $p < 0.001$) experience were positive related with eWOM ($\Delta R^2 = 0.44$), supporting H4a and H4b. This means the higher feeling of flow and aesthetic experience on brand-related content, the more positive toward the marketing effectiveness. (see Table 2 for details).

5. Conclusions

5.1 Discussion

Brands can cultivate more robust consumer relationships by enhancing the value of their online content, a facet that can be refined through improved visual presentation. As previous research suggests, color plays a pivotal element, acting as a cue for users to shape their perceptions of information sources, thereby consistently influencing attention and attitude (Chiu *et al.*, 2017; Zailskaitė-Jakštė *et al.*, 2017). Despite the well-established understanding of

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color effects in online environments, limited research has specifically delved into the impact of photo and image colors on Instagram branding. The present study endeavors to fill this research gap by investigating the color congruence between the background and themes of brand posts on Instagram, drawing insights from a dataset of 280 valid samples. The key results of this study are as follows.

Firstly, the findings of this study revealed that brand layouts and posts on Instagram are more likely to elicit the flow experience through color congruence as opposed to color incongruence, aligning with the results of a previous study (Wu *et al.*, 2016). Moreover, congruent information is more likely to evoke an aesthetic experience, in line with the findings of an earlier investigation (Hall and Hanna, 2004). Specifically, this phenomenon can be attributed to the notion that congruent colors in photos facilitate smooth cognitive processing for viewers, thereby fostering a positive flow and aesthetic experience. Consequently, brands that integrate congruent colors into their layouts and posts on Instagram have the ability to capture consumers' attention, intensify their interest, and consequently elicit more favorable response rates.

Secondly, in alignment with prior research conducted by Aggarwal and McGill (2012) and Chiu (2022), the results emphasize that flow states have the potential to enhance users' evaluations of brand attitude and eWOM. Notably, this study unveils a significant impact of flow and aesthetic states on brand attitude and eWOM within brand posts, contrary to findings in previous studies. This deviation may be attributed to the proposition that flow and aesthetic states play a crucial role in augmenting users' overall experience and attitude toward brand posts, enabling a more profound perception of content. As users identify the content as significant, they may develop a more positive attitude and an increased intent to share it, aligning with the mechanisms proposed by Storbeck and Clore (2008) and Yu and Ko (2021).

Thirdly, the results not only validated the premise that flow states enhance visit intention, consistent with prior research (Yang *et al.*, 2020), but also revealed that flow states can significantly and effectively increase visit intention. Beyond the realms of traditional marketing (Poels and Dewitte, 2006), this study highlights the substantial impact of flow experience on marketing effectiveness within the context of Instagram. By examining the role of experiential value through indirect visual representation in the form of brand layouts and posts on Instagram, this study delves into the significance of evoking flow and aesthetic experiences, subsequently influencing behavioral responses.

The significance of this study is underscored by its contributions to the existing literature, stemming from the analysis and classification of real brand cases and the empirical testing of effects. This study not only applied the congruence effect but also broadened its scope to encompass novel and crucial aspects of visual content. From a practical standpoint, our research furnishes valuable foundational data regarding the tangible impact on marketing strategy employing brand experience and social media.

Table 2.
The impact of processing fluency on marketing effectiveness

Dependent variable	Independent variable	β	t	p	ΔR^2
Brand attitude	Flow experience	0.60	8.62	<0.001*	0.43
	Aesthetic experience	0.32	4.86	<0.001*	
Visit intention	Flow experience	0.43	6.54	<0.001*	0.49
	Aesthetic experience	0.32	4.87	<0.001*	
eWOM	Flow experience	0.35	5.08	<0.001*	0.44
	Aesthetic experience	0.35	5.07	<0.001*	

Note(s): β = beta coefficients; t = t-value; p = p-value
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5.2 Theoretical implications

Compared with previous studies, this research contributed to academia in several noteworthy aspects. Firstly, contemporary studies have delved into various sources of flow experience, encompassing factors such as high visual clarity in reading materials (Ettis, 2017), repeated exposure (Hyun *et al.*, 2022; Nunes *et al.*, 2015), and perceived visual congruency (Cuevas *et al.*, 2021; Sohn, 2017). However, prior to this study, there has been a dearth of exploration into how color might induce the congruence effect specifically in the context of Instagram brand posts. By extending the perspective on color congruence (Lee and Kim, 2022; Wong *et al.*, 2019), this study pioneers an investigation into the impact of color on the Instagram platform, representing an initial foray in this research direction by scrutinizing the effect of color congruence on users' perceptions of brand posts. Additionally, the study explored how color congruence may trigger aesthetic experiences in relation to brand posts, offering a nuanced understanding of the interplay between color and aesthetic perceptions.

Secondly, the experimental outcomes of this study revealed that users' marketing effectiveness can be influenced by the color congruence heuristic between themes and background colors in brand posts. Notably, the study employed blue as the primary color in the experiment, aligning with previous research, and identified a significant impact of color congruence on the results (Chiu *et al.*, 2017; Seifert and Chattaraman, 2020). Furthermore, this investigation expands upon Crutzen *et al.* (2012) and Kim *et al.* (2020) proposition regarding congruence, delving deeper into the influence of color congruence on both flow and aesthetic experiences. The findings affirm that consistently vibrant colors contribute to the generation of flow and aesthetic experiences.

Thirdly, prior research has explored the impact of flow experience on marketing effectiveness using various dimensions and contexts (Cuevas *et al.*, 2021; Yang *et al.*, 2020). This study extended this perspective by specifically examining the influence of flow and aesthetic experience on Instagram posts. The findings indicate that both flow and aesthetic experiences significantly affect brand attitude, visit intention, and eWOM. In sum, this study contributed to the existing body of literature on social media marketing (Yu and Ko, 2021), highlighting the nuanced role of color as a complex construct operating within diverse dimensions, enhancing the perception of brand-related posts. The outcomes of this study can serve as valuable references for future research endeavors.

5.3 Practical implications

This study provides foundational insights into the tangible impact of color on brand posts within Instagram marketing strategies. Initially, consumers leverage available marketing cues, encompassing atmospherics and visual elements on Instagram, to form judgments about brand attitudes and establish purchase intentions. Brands managing their official Instagram accounts should pay close attention to the overall color scheme of the layout. This not only minimizes cognitive processing time and prevents clutter but also aids consumers in promptly recognizing their brand. Particularly on visual-centric platforms like Instagram, the capacity to preserve tonal values in post editing enables brands to maintain a consistent color style, thereby facilitating more efficient account management.

Secondly, not all marketing stimuli exert equal effectiveness, warranting increased investment in stimuli likely to evoke users' flow and aesthetic experiences. This study highlights the potent impact of visual communication, demonstrating that congruence and incongruence in colors can induce flow states and aesthetic experience within the contexts of brand posts/layouts. Online users heavily rely on color cues and processing states to discern perceived value across different brands. Brands should strategically utilize colors in their layouts/posts on Instagram to elicit users' flow and aesthetic experiences, consequently

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enhancing consumers' attitudes towards the brands. For instance, brands can incorporate their distinctive color as the primary hue, connecting more deeply with their brand image and augmenting perceived value toward the product.

Thirdly, marketers should meticulously design and deploy color cues in their social media strategies to evoke desired emotional reactions and behaviors. With heightened flow and aesthetic experiences, marketing effectiveness is likely to follow suit. Importantly, consumers often form brand impressions based on the colors used. Maintaining color congruence with the brand's attributes on official accounts can enhance consumer satisfaction and elevate brand evaluations. The findings suggest that brands can devise more effective content strategies by leveraging color cues to stimulate consumers' flow and aesthetic experiences.

5.4 Limitations and future research

This study has certain limitations and points to avenues for future research. Firstly, the distinctive characteristics of Instagram may have influenced our results. Replicating this model on alternative online platforms would provide a more comprehensive understanding of the color effect on marketing campaigns. Additionally, exploring color congruence across diverse cultural groups is essential for a more nuanced perspective, considering the influence of culture.

Secondly, our focus was on the sports brand Adidas Originals, and the generalizability of our findings to other industries remains unexplored. Future research should investigate the proposed interaction effect with different brands and consider potential gender differences in brand responses to visual cues.

Lastly, this study centered on the Adidas Originals' brand colors, primarily blue and white. Exploring the impact of other colors on user perceptions and attitudes is an area for further investigation. Additionally, understanding how varying levels of color diversity fluency may influence the demonstrated interaction effect requires more research. The diverse meanings associated with color hues also indicate a need for refined aesthetic values, potentially leading to further segmentation in this domain.

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(The Appendix follows overleaf)

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Appendix

Time distortion
TD1. Time appears to go by very quickly when I am using this Instagram account
TD2. Sometimes I lose track of time when I am using this Instagram account
TD3. Time flies when I am using this Instagram account
TD4. Most times when I get on to this Instagram account, I end up spending more time that I had planned
TD5. I Often spend more time on this Instagram account than I had intended.
Immersion
Fl1. While using this Instagram account, I am able to block out most other distractions
F12. While using this Instagram account, I am absorbed in what I am doing
F13. While on this Instagram account, I am immersed in the task I am performing
F14. When on this Instagram account, I get distracted by other attentions very easily
F15. While on this Instagram account, my attention does not get diverted very easily
Enjoyment
HE1. I have fun interacting with this Instagram account
HE2. Using this Instagram account provides me with a lot of enjoyment
HE3. I enjoy using this Instagram account
HE4. Using this Instagram account bores me
Control
C01. When using this Instagram account, I feel in control
C02. I feel that I have no control over my interaction with this Instagram account
C03. This Instagram account allows me to control my computer interaction
Aesthetics experience
EA1. This Instagram account is pleasing to look at
EA2. I like the look and feel of this Instagram account
EA3. This Instagram account is visually appealing
EA4. The visual design of this Instagram account is attractive
EA5. This Instagram account looks pretty
Brand attitude
BA1. The brand was (negative-positive)
BA2. The brand was (unattractive-attractive)
BA3. The brand was (unfavorable-favorable)
BA4. The brand was (bad-good)
BA5. The brand was (unimportant-important)
Intention to revisit
IR1. I intend to revisit this Instagram account
IR2. I want to visit this Instagram account again
IR3. I have a revisit intention for this Instagram account
IR4. I will visit this Instagram account again
eWOM
EW1. I will inform others of the brand
EW2. I will share (like, tags, comments, etc.) information about the brand with others
EW3. I will positively inform others of the brand
EW4. I will recommend the brand to others

Table A1. The specific items in each scale in this study

Source(s): Created by author

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